

The Ringdown Register

CURRICULUM VITAE OF ANUJ KANKANI · GRAVITATIONAL-WAVE PHYSICS

Gravitational Waveform Modeling — Numerical Relativity — General Relativistic Particle-in-Cell Simulation

VOL. I — No. 1

MORGANTOWN, WEST VIRGINIA — SUNDAY, AUGUST 16, 2026

PRICE FIVE CENTS

WEST VIRGINIA UNIVERSITY · DEPARTMENT OF PHYSICS & ASTRONOMY

KANKANI MODELS BLACK-HOLE MERGERS AND THE PLASMA THAT SURROUNDS THEM

SEVEN PUBLICATIONS, SIX OF THEM FIRST-AUTHORED, WITH THREE
MORE IN PREPARATION

PRIMARY DEVELOPER OF gwBOB — CONTRIBUTOR TO THE EINSTEIN TOOLKIT, TO ENTITY
AND TO NRPY — CO-DEVELOPER OF LISAWAVE-JAX

Doctoral Candidate at West Virginia University — Expected 2027 — Advisor, Sean T. McWilliams

ANUJ KANKANI is a doctoral candidate in the Department of Physics & Astronomy at West Virginia University, Morgantown, under Sean T. McWilliams, expecting to graduate in 2027.

He works on gravitational waveform modeling, numerical relativity, and General Relativistic Particle-in-Cell (GRPIC) simulations. He took his BS in Physics (Honors) at Texas A&M University in May 2021, with

minors in Computer Science and Astrophysics.

Of his seven publications, four are submitted and under review; the three on the direct wave and the BOB waveform model are set out at the foot of this page. Three others are refereed and in print — two in *Physical Review D*, one in *Physical Review E* — and appear on Page Two beside the fourth submission. Six of the seven carry his name first.

He is the primary developer of gwBOB, an open-source Python package implementing the Backwards One Body waveform model, contributes to the Einstein Toolkit, to Entity and to NRPY, and co-develops LisaWave-JAX. All five are on Page Three.

He is a member of the LISA Consortium, in its waveform and astrophysics working group. His engagements are on Page Two; his service and outreach, on Page Three.

THIS OFFICE

ANUJ KANKANI

Department of Physics &
Astronomy
West Virginia University
Morgantown, West Virginia

POST
anuj.kankani@mail.wvu.edu
WEB
anujkankani.github.io
ORCID
0000-0002-7422-9137
CODE
github.com/AnujKankani
INSPIRE
inspirehep.net/authors/2774144

TERMS OF STUDY

WEST VIRGINIA UNIVERSITY

PhD in Physics — in progress
Expected graduation 2027. Advisor:
Sean T. McWilliams. Grade-point
average 3.9 / 4.0.

TEXAS A&M UNIVERSITY

BS in Physics (Honors)
College Station, Texas. August
2017 – May 2021. Minors in Com-
puter Science and Astrophysics.
Grade-point average 3.84 / 4.0.

HONOURS CONFERRED

RUBY DISTINGUISHED DOCTORAL FELLOW 2021– 2024

Awarded to outstanding doctoral
students in STEM fields at West
Virginia University.

NASA WV SPACE GRANT GRADUATE FELLOWSHIP 2024–2025

For the proposal “Modeling Black
Hole Mergers with a Dynamical
Background Spacetime.”

MOHINDAR SEEHRA RESEARCH AWARD 2025

Departmental award recognizing
outstanding research efforts in
Physics.

LATEST DISPATCHES

Three papers submitted and under review. Refereed work in print appears on Page Two.

UNDER REVIEW · 2026

THE DIRECT WAVE IS NOT A MEANINGFUL TEST OF HORIZON PROPERTIES

SUBMITTED TO *PHYSICAL
REVIEW LETTERS*

Anuj Kankani and Sean T.
McWilliams

ARXIV:2607.02380

**

First author · Not yet refereed

UNDER REVIEW · 2026

DIRECT WAVES IN BLACK-HOLE BINARY MERGERS: INSIGHTS FROM THE BACKWARDS ONE BODY MODEL

SUBMITTED TO *PHYSICAL
REVIEW D*

Anuj Kankani and Sean T.
McWilliams

ARXIV:2603.15474

**

First author · Not yet refereed

UNDER REVIEW · 2025

BOB THE (WAVEFORM) BUILDER: OPTIMIZING ANALYTICAL BLACK- HOLE BINARY MERGER WAVEFORMS

SUBMITTED TO *PHYSICAL
REVIEW D*

Anuj Kankani and Sean T.
McWilliams

ARXIV:2510.25012

**

First author · Not yet refereed

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Three refereed in print,
one more submitted

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THE PUBLISHED RECORD

NOTICE TO THE READER. Each entry carries its standing in the rule above it. The first three are refereed and in print, with volume and article numbers; the fourth, like the three on Page One, is submitted and not yet refereed.

REFEREED — IN PRINT

MODELING RELATIVE PEAK TIMES OF GRAVITATIONAL WAVE HARMONICS

PHYS. REV. D 112, 124051 (2025)
Anuj Kankani and Sean T. McWilliams
ARXIV:2506.17492 · FIRST AUTHOR

REFEREED — IN PRINT

TESTING THE BOUNDARY-TO-BOUND CORRESPONDENCE WITH NUMERICAL RELATIVITY

PHYS. REV. D 110, 064033 (2024)
Anuj Kankani and Sean T. McWilliams
ARXIV:2404.03607 · FIRST AUTHOR

REFEREED — IN PRINT

OPTIMIZATION AND BENCHMARKING OF THE THERMAL CYCLING ALGORITHM

PHYS. REV. E 104(3), 035302 (2021)
A. Barzegar, A. Kankani, S. Mandrà and H. G. Katzgraber
SECOND AUTHOR

SUBMITTED — UNDER REVIEW

GWBOB: A PYTHON PACKAGE FOR ANALYTICAL MERGER-RINGDOWN GRAVITATIONAL WAVEFORMS

SUBMITTED TO *Journal of Open Source Software*
Anuj Kankani, Angel Morales, Suchindram Dasgupta and Sean T. McWilliams
FIRST AUTHOR

IN PREPARATION. Three further papers are presently in preparation. Titles and journals are not yet settled; particulars will follow in a later edition.

ENGAGEMENTS AND APPEARANCES

Twelve meetings and workshops — nine talks, two posters, one attended.

2026 PLASMAS AROUND BLACK HOLES AND NEUTRON STARS (SCEECs)
POSTER
“Driving Temperature Anisotropies in the Jets of Black Holes”

2026 16TH INTERNATIONAL LISA SYMPOSIUM
CONTRIBUTED TALK
“BOB the (Waveform) Builder + Direct Waves”

2026 APS GLOBAL PHYSICS SUMMIT
CONTRIBUTED TALK
“BOB the (Waveform) Builder: Optimizing Analytical Black-Hole Binary Merger Waveforms”

2026 CENTER FOR GRAVITATIONAL WAVES AND COSMOLOGY SYMPOSIUM
CONTRIBUTED TALK
“Modeling Extreme Gravity: From Relativistic Plasmas to Gravitational Waves”

2025 PROSPECTS IN THEORETICAL PHYSICS — GRAVITATIONAL WAVES FROM THEORY TO OBSERVATION
ATTENDED

2025 APS GLOBAL PHYSICS SUMMIT
CONTRIBUTED TALK
“Merger-Ringdown Modeling of Gravitational Waves Beyond the Dominant Mode”

2025 AAS WINTER MEETING
CONTRIBUTED TALK
“Understanding the Origins of Black Hole Superkicks”

2024 NORTH AMERICAN EINSTEIN TOOLKIT WORKSHOP
THIRTY-MINUTE TALK
“Understanding the Origins of Black Hole Superkicks”

2024 PENN STATE NEIGHBORHOOD WORKSHOP
CONTRIBUTED TALK
“Understanding the Origins of Black Hole Superkicks”

2024 APS APRIL MEETING
CONTRIBUTED TALK
“Testing the Boundary to Bound States Dictionary with Numerical Relativity”

2023 AAS WINTER MEETING
POSTER
“Effect of Baryons on the Dynamical-Friction Timescale of Merging Dark Matter Halos”

2022 APS APRIL MEETING
CONTRIBUTED TALK
“Effect of Baryons on the Dynamical-Friction Timescale of Merging Dark Matter Halos”

SOFTWARE AND CODE

*Five open-source projects — primary developer of one, co-developer of another, contributor to three.***gwBOB****PRIMARY DEVELOPER**github.com/AnujKankani/BackwardsOneBody

Open-source, user-friendly Python package implementing the Backwards One Body waveform model, with tools for analysis and NR comparison.

LisaWave-JAX**CO-DEVELOPER — IN DEVELOPMENT**

JAX-based LISA data-analysis tools for massive black holes and galactic binaries.

Entity**CONTRIBUTOR**entity-toolkit.github.io/wiki

Kokkos-based General Relativistic Particle-in-Cell code for plasma astrophysics.

Einstein Toolkit**CONTRIBUTOR**einstein toolkit.org

Open-source high-performance-computing codebase for numerical relativity.

NRPy**CONTRIBUTOR**github.com/nrpy/nrpy

Python-based symbolic code-generation toolkit for relativistic astrophysics.

CIVIC NOTICES

*Consortium membership, departmental service, outreach and community work.***LISA CONSORTIUM**

Member of the waveform and astrophysics working group.

ASTRONOMY JOURNAL CLUB — ORGANIZER

Organized and led the department's weekly journal club, facilitating discussions on recent research in astronomy and astrophysics.

COMMUNITY SERVICE — 200 HOURS

Two hundred hours of community service completed as a graduate student, focused on environmental restoration and STEM outreach.

PHYSICS AND ASTRONOMY GRADUATE STUDENT ORGANIZATION

Regularly involved in STEM outreach such as Physics Demo Day and the pumpkin drop contest.

PHYSICS OUTREACH

Talks to middle- and high-school groups about black holes and gravitational waves; regular volunteer at physics outreach events such as Physics Demo Day; writes articles on astrophysics for local organizations.

TERMS OF CORRESPONDENCE

BY POST

Department of Physics & Astronomy
West Virginia University
Morgantown, West Virginia

BY WIRE

anuj.kankani@mail.wvu.edu
anujkankani.github.io

IN THE INDEXES

ORCID 0000-0002-7422-9137
github.com/AnujKankani
inspirehep.net/authors/2774144

COLOPHON AND STATEMENT OF THE PRESS

The Ringdown Register is not a newspaper. It is the curriculum vitæ of Anuj Kankani, set in the manner of a broadsheet front page of the eighteen-nineties. The title is invented; no publication of that name exists, and none is imitated. Every fact printed in these columns is drawn from that curriculum vitæ, and nothing has been added for effect — there are no quotations, no correspondents, no reports, and no

assessment of the work beyond what its author has stated. The volume number, the price line and the ornaments are costume; the dateline is the day of setting. Set in Old English Text MT for the nameplate, Bodoni MT for headlines, Engravers MT for standing heads, and Times New Roman for text and decks. Composed in HTML and printed to PDF. Morgantown, West Virginia, August 2026.